

RESEARCH
INTERESTS

My research spans the interdisciplinary fields of machine learning and natural language processing, with a particular focus on addressing societal challenges through computational methods. I aim to develop tools and apply techniques to uncover complex patterns that can be leveraged to tackle pressing social issues.

EDUCATION

- Sabanci University**, Istanbul, Turkey *Sept. 2024 - Present*
PhD, **Computer Science and Engineering**
Advisor: **Asst. Prof. Onur Varol**
- Sabanci University**, Istanbul, Turkey *Feb. 2022 - May 2024*
M.Sc., **Computer Science and Engineering**
Thesis: Developing Turkish Language Models on Social Media
Advisor: **Asst. Prof. Onur Varol**
- University Of Tabriz**, Tabriz, Iran *Sept. 2017 - Sept. 2021*
B.Sc., **Computer Science and Engineering**

PUBLICATIONS
GOOGLE SCHOLAR
CITATIONS: 86
H-INDEX: 4
I10-INDEX: 2

Journal Articles

- J.4 **Najafi, A.**, Iannucci, L., Kivelä, M., & Varol, O. (2026). **“Social Media Data Toolkit: Standardization and Anonymization of Social Network Datasets.”** arXiv preprint arXiv:2604.27710. [↗](#)
- J.3 **Najafi, A.**, & Varol, O. (2024). **“Turkishbertweet: Fast and reliable large language model for social media analysis.”** Expert Systems with Applications, 255, 124737. [↗](#)
- J.2 **Najafi, A.**, Mugurtay, N., Zouzou, Y., Demirci, E., Demirkiran, S., Karadeniz, H. A., & Varol, O. (2024). **“First public dataset to study 2023 Turkish general election.”** Scientific Reports, 14(1), 8794. [↗](#)
- J.1 **Najafi, A.**, Gholipour-Shilabin, A., Dehkharghani, R., Mohammadpur-Fard, A., & Asgari-Chenaghlu, M. (2023). **“ComStreamClust: a communicative multi-agent approach to text clustering in streaming data.”** Annals of data science, 10(6), 1583-1605. [↗](#)

Refereed Conference Proceedings

- C.2 Yener, A. Ö., **Najafi, A.**, & Varol, O. (2024, May). **“City-level User Location Detection from Social Media and Analysis of Political Participation.”** In 2024 32nd Signal Processing and Communications Applications Conference (SIU) (pp. 1-4). IEEE. [↗](#)
- C.1 **Najafi, A.**, & Varol, O. (2024, March). **“VRLLab at HSD-2Lang 2024: Turkish hate speech detection online with TurkishBERTweet.”** In Proceedings of the 7th Workshop on Challenges and Applications of Automated Extraction of Socio-political Events from Text (CASE 2024) (pp. 185-189). [↗](#)

CONDUCTED
RESEARCH AND
ACADEMIC
EXPERIENCE

- VRL LAB**, Istanbul, Turkey *Sept. 2022 - Present*
- **Eclipse Horizon Project** funded by the European Union
 - * Researcher developing the Network Science Tool Package to analyze and mitigate hate speech across social media platforms.
 - **CON-NET** under CHIST-ERA call funded by EU and TUBITAK
 - * Developed the Social Media Data Toolkit (SMDT) for standardizing and anonymizing social network datasets, resulting in the J.4 pre-print publication.

– **Turkish Election Dataset**

- * Analyzed the Retweet Speed of specific Turkish Politicians for different time spans of election day.
- * Conducted network analysis of politicians on Twitter using NetworkX and Gephi.
- * Analyzed the followers of politicians to identify social bots using Botometer.

– **TurkishBERTweet**

- * Benchmarked various models on downstream tasks to evaluate comparative performance.
- * Achieved almost 10% significant improvement in the F1-Score of Text Classification Tasks.
- * Developed classification pipelines for Sentiment Analysis and Hate Speech Detection Tasks.
- * Used Google TPUs to train TurkishBERTweet from scratch.
- * Trained BPE tokenizer from scratch on 894M Turkish Twitter Data.

INVITED
LECTURER AND
TALKS

1. Presented the paper “VRLLab at HSD-2Lang 2024: Turkish Hate Speech Detection Online with TurkishBERTweet” at [EACL”2024](#) , Malta, March 2024.
2. Presented the paper “First Public Dataset to Study the 2023 Turkish General Election” at the [Turkey Computational Social Sciences Conference \(TCSS\) 2023](#) , Koc University, Turkey.
3. Invited lecturer for the talk “Applications of Large Language Models for Social Scientists” in the Political Science course at Sabanci University, Turkey, under the supervision of [Dr. Fatih Serkant Adiguzel](#), Fall 2023.
4. Invited lecturer for the talk “Natural Language Processing for Social Scientists” at the [Summer Institutes in Computational Social Science \(SICSS\)](#) , Ozyegin University, Turkey, Summer 2023.

TEACHING
EXPERIENCE

Sabancı University - Faculty of Engineering and Natural Sciences

CS210: Introduction to Data Science

Spring 2023, Fall 2023

CS412: Machine Learning

Fall 2024, Spring 2024, Fall 2022

CS201: Introduction to Computing

Spring 2022

Held weekly office hours and recitation sessions to address course-related questions.
Graded exams and assignments.

COMPUTER AND
LANGUAGE
SKILLS

Skills:

Programming Languages: Python, C++, Java

Frameworks: PyTorch, TensorFlow, Keras, Flax, Django

Libraries: Matplotlib, Seaborn, Pandas, NumPy, Gephi, NetworkX, OpenCV, JAX, Docker

Databases: SQL, NoSQL (Redis)

Familiar with GCP, AWS, $\text{\LaTeX}$, HTML, CSS, JS

Languages: English (Fluent), Persian (Native), Azerbaijani (Native), Turkish (Fluent)

HONORS AND
AWARDS

[Hate Speech Detection in Turkish - EACL”2024](#)

Jan. 2024

Earned 3rd Rank among 11 participating teams.

[TUBITAK Scholarship](#)

Aug. 2023 - Present

The Scientific and Technological Research Council of Turkey, Turkey

Awarded a full tuition waiver and monthly stipend as a researcher on a TUBITAK-funded project.

Sabancı University Scholarship

Feb. 2022 - Aug. 2023

Sabancı University, Istanbul, Turkey

Awarded a 100% tuition scholarship and a monthly stipend as a Teaching Assistant.